

Math 231B HW 1

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Problem 1

Etingof Problem Set 7.1:

Let $R \subset \mathbb{R}^n$ be given by

$$R = \{\pm e_i, \pm 2e_i \mid 1 \leq i \leq n\} \cup \{\pm e_i \pm e_j \mid 1 \leq i, j \leq n, i \neq j\} \quad (1.1)$$

where e_i is the standard basis in $\mathbb{R}^n$. Show that R is a non-reduced root system.

Solution: First, it's clear that R spans the whole space, since all the standard basis elements are contained in R .

Now, given any $\alpha, \beta \in R$ (WLOG, say they're positive components in R by factoring out the -1 , we'll test it for the e_i 's, $2e_i$'s, and $e_i \pm e_j$'s), we want to show the property $\frac{2(\alpha, \beta)}{(\alpha, \alpha)} \in \mathbb{Z}$. Here are some cases:

- If both are e_i, e_j , one has $\frac{2(e_j, e_i)}{(e_i, e_i)} = 2\delta_{ij} \in \mathbb{Z}$.
- If one is e_i while the other is $2e_j$, one has $\frac{2(e_i, 2e_j)}{(e_i, e_i)} = 4\delta_{ij} \in \mathbb{Z}$; for the other order, one also has $\frac{2(2e_j, e_i)}{(2e_j, 2e_j)} = \frac{4\delta_{ij}}{4} = \delta_{ij} \in \mathbb{Z}$.
- If one is e_i and the other is $e_j \pm e_k$, one has $\frac{2(e_i, e_j \pm e_k)}{(e_i, e_i)} = 2(\delta_{ij} \pm \delta_{ik}) \in \mathbb{Z}$; for the other order, one also has $\frac{2(e_j \pm e_k, e_i)}{(e_j \pm e_k, e_j \pm e_k)} = \left(2 \frac{\delta_{ij} \pm \delta_{ik}}{2}\right) = \delta_{ij} \pm \delta_{ik} \in \mathbb{Z}$.
- If both are $2e_i, 2e_j$, one has $\frac{2(2e_i, 2e_j)}{(2e_i, 2e_i)} = \frac{8\delta_{ij}}{4} = 2\delta_{ij} \in \mathbb{Z}$.
- If one is $2e_i$ while the other is $e_j \pm e_k$, one has $\frac{2(2e_i, e_j \pm e_k)}{(2e_i, 2e_i)} = \frac{4(\delta_{ij} \pm \delta_{ik})}{4} = \delta_{ij} \pm \delta_{ik} \in \mathbb{Z}$; for the other order, one also has $\frac{2(e_j \pm e_k, 2e_i)}{(e_j \pm e_k, e_j \pm e_k)} = \frac{4(\delta_{ij} \pm \delta_{ik})}{2} = 2(\delta_{ij} \pm \delta_{ik}) \in \mathbb{Z}$.
- Finally, if both are $e_i \pm_{ij} e_j, e_k \pm_{kl} e_l$, one has $\frac{2(e_i \pm_{ij} e_j, e_k \pm_{kl} e_l)}{(e_i \pm_{ij} e_j, e_i \pm_{ij} e_j)} = \frac{2(\delta_{ik} \pm_{ij} \delta_{jk} \pm_{kl} \delta_{il} + (\pm_{ij} 1)(\pm_{kl} 1)\delta_{jl})}{2} = (\delta_{ik} \pm_{ij} \delta_{jk} \pm_{kl} \delta_{il} + (\pm_{ij} 1)(\pm_{kl} 1)\delta_{jl}) \in \mathbb{Z}$.

Then, the last condition is regarding the reflections (which the coefficients will be coming from the above calculation):

- Given $s_{e_i}(e_j) = e_j - 2\delta_{ij}e_i$, if $i \neq j$ this is $e_j \in R$; else if $i = j$ this is $-e_i \in R$.
- Given $s_{e_i}(2e_j) = 2e_j - 4\delta_{ij}e_i$, if $i \neq j$ this is $2e_j \in R$; else if $i = j$ this is $-2e_i \in R$.

When reversing the order, $s_{2e_j}(e_i) = e_i - \delta_{ij}2e_j$, if $i \neq j$ this is $e_i \in R$; else if $i = j$ this is $-e_i \in R$.

- Given $s_{e_i}(e_j \pm e_k) = (e_j \pm e_k) - 2(\delta_{ij} \pm \delta_{ik})e_i$, for $i = j$ (which $i \neq k$) this becomes $-e_i \pm e_k \in R$; for $i = k$ (which $i \neq j$) this becomes $e_j \mp e_i \in R$; else if $i \neq j$ and $i \neq k$, it's just $e_j \pm e_k \in R$.

When reversing the order, $s_{e_j \pm e_k}(e_i) = e_i - (\delta_{ij} \pm \delta_{ik})(e_j \pm e_k)$, for $i = j$ (which $i \neq k$) this is $\mp e_k \in R$; if $i = k$ (which $i \neq j$) this is $\mp e_j \in R$; else if $i \neq j$ and $i \neq k$, this is $e_i \in R$.

- Given $s_{2e_i}(2e_j) = 2e_j - 2\delta_{ij}(2e_i)$, if $i \neq j$ this is $2e_j \in R$; else if $i = j$ this is $-2e_i \in R$.
- Given $s_{2e_i}(e_j \pm e_k) = (e_j \pm e_k) - (\delta_{ij} \pm \delta_{ik})2e_i$, if $i = j$ (which $i \neq k$) this is $-e_i \pm e_k \in R$; if $i = k$ (which $i \neq j$) this is $e_j \in R$; else if $i \neq j$ and $i \neq k$ this is $e_j \pm e_k \in R$.

When reversing the order, $s_{e_j \pm e_k}(2e_i) = 2e_i - 2(\delta_{ij} \pm \delta_{ik})(e_j \pm e_k)$, if $i = j$ (which $i \neq k$) this is $\mp 2e_k \in R$; if $i = k$ (which $i \neq j$) this is $\mp 2e_j \in R$; else if $i \neq j$ and $i \neq k$ this is $2e_i \in R$.

- Given $s_{e_i \pm_{ij} e_j}(e_k \pm_{kl} e_l) = (e_k \pm_{kl} e_l) - (\delta_{ik} \pm_{ij} \delta_{jk} \pm_{kl} \delta_{il} + (\pm_{ij}1)(\pm_{kl}1)\delta_{jl})(e_i \pm_{ij} e_j)$ (yep this term is diabolical...). If $\{i, j, k, l\}$ are four distinct indices, this reduces to $e_k \pm_{kl} e_l \in R$; if they form only three distinct indices (say $i = k$ up to some reordering and $\pm$ sign) then $i \neq j$, $j \neq k$ and $i, j \neq l$, so the term reduces to $\pm_{kl} e_l \mp_{ij} e_j \in R$; finally, if they form only two distinct indices (say $i = k$, $j = l$ up to reordering and $\pm$ signs), if $\pm_{ij}$ and $\pm_{kl}$ have the same sign, the term reduces to $-e_i \mp e_j \in R$, else if $\pm_{ij}$ and $\pm_{kl}$ have different signs, this reduces to $e_k \pm_{kl} e_l \in R$.

So, all possible reflection combinations of the roots (up to signs and reorderings) are still in R , showing it satisfies the root system axioms.

Then, since $e_i \in R$ satisfies $2e_i \in R$, this shows that the root system is non-reduced.

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Problem 2

Etingof Problem Set 7.2:

- (1) Let $R \subset E$ be a root system. Show that the set

$$R^\vee = \{\alpha^\vee \mid \alpha \in R\} \subset E^* \quad (2.1)$$

where $\alpha^\vee \in E^*$ is the coroot corresponding to α is also a root system. It is usually called the *dual root system* of R .

- (2) Let $\Pi = \{\alpha_1, \dots, \alpha_r\} \subset R$ be the set of simple roots. Show that the set $\Pi^\vee = \{\alpha_1^\vee, \dots, \alpha_r^\vee\} \subset R^\vee$ is the set of simple roots of $R^\vee$.

(Note: this is not completely trivial, as $\alpha \mapsto \alpha^\vee$ is not a linear map.)

Solution: Given that E is an $\mathbb{R}$ -inner product space, hence the linear map $E \rightarrow E^*$ by $v \mapsto (v, _)$ is an isomorphism. Which, given any $\varphi, \psi \in E^*$, say $\varphi = (v, _)$ and $\psi = (u, _)$ for $v, u \in E$, define an inner product $(_, _)_* : E^* \times E^* \rightarrow \mathbb{R}$ by $(\varphi, \psi)_* = (v, u)$.

In particular, for any root $\alpha \in R$, the coroot $\alpha^\vee(_) = \frac{2(\alpha, _)}{(\alpha, \alpha)} = \left(2\frac{\alpha}{(\alpha, \alpha)}, _ \right)$.

- (1) First, to show $R^\vee$ is a root system of E^* , notice that since the root system R spans E , while the assignment $\alpha \mapsto (\alpha, _)$ defines an isomorphism $E \simeq E^*$, then the collection $R' = \{(\alpha, _) \mid \alpha \in R\} \subset E^*$ is the image of R under the assignment, which spans E^* .

Since every $\alpha \in R$ is nonzero, then by positive definiteness, $(\alpha, _) \neq 0$ and $(\alpha, \alpha) > 0$. Hence, up to rescaling, one has $R^\vee = \{\alpha^\vee \mid \alpha \in R\} = \left\{ \frac{2}{(\alpha, \alpha)}(\alpha, _) \mid \alpha \in R \right\}$, with each element being a rescale of a unique element of R' (and for every element in R' , say $(\alpha, _)$, its rescale $\frac{2}{(\alpha, \alpha)}(\alpha, _)$ is in $R^\vee$). Hence, one has $R^\vee$ spanning E^* (since R' does).

Second, for any $\alpha, \beta \in R$, the corresponding coroot $\alpha^\vee = \left(2\frac{\alpha}{(\alpha, \alpha)}, _ \right)$ and $\beta^\vee = \left(2\frac{\beta}{(\beta, \beta)}, _ \right)$. So, by the above definition of the induced inner product on E^* , one has the following:

$$\begin{aligned} (\alpha^\vee, \beta^\vee)_* &= \left(2\frac{\alpha}{(\alpha, \alpha)}, 2\frac{\beta}{(\beta, \beta)}\right) = \frac{4(\alpha, \beta)}{(\alpha, \alpha)(\beta, \beta)} \\ (\alpha^\vee, \alpha^\vee)_* &= \left(2\frac{\alpha}{(\alpha, \alpha)}, 2\frac{\alpha}{(\alpha, \alpha)}\right) = \frac{4}{(\alpha, \alpha)} \end{aligned} \quad (2.2)$$

Hence, the value $n_{\beta^\vee, \alpha^\vee}$ is given as follow:

$$n_{\beta^\vee, \alpha^\vee} = \frac{2(\alpha^\vee, \beta^\vee)}{(\alpha^\vee, \alpha^\vee)} = \frac{2 \cdot 4(\alpha, \beta)}{(\alpha, \alpha)(\beta, \beta)} \cdot \frac{(\alpha, \alpha)}{4} = \frac{2(\alpha, \beta)}{(\beta, \beta)} = n_{\alpha, \beta} \in \mathbb{Z} \quad (2.3)$$

Now, regarding the reflections, one has the following:

$$\begin{aligned} s_{\alpha^\vee}(\beta^\vee) &= \beta^\vee - n_{\beta^\vee, \alpha^\vee} \alpha^\vee = \left(2\frac{\beta}{(\beta, \beta)}, _ \right) - \frac{2(\alpha, \beta)}{(\beta, \beta)} \left(2\frac{\alpha}{(\alpha, \alpha)}, _ \right) \\ &= \frac{2}{(\beta, \beta)} \left(\beta - \frac{2(\alpha, \beta)}{(\alpha, \alpha)} \alpha, _ \right) = \frac{2}{(\beta, \beta)} (\beta - n_{\beta, \alpha} \alpha, _) \\ &= \frac{2}{(\beta, \beta)} (s_\alpha(\beta), _) \end{aligned} \quad (2.4)$$

Lastly, remember that for any $\alpha, \beta \in R$, one has $(\beta, \beta) = (s_\alpha(\beta), s_\alpha(\beta))$, so it reduces to the following:

$$s_{\alpha^\vee}(\beta^\vee) = \frac{2}{(s_\alpha(\beta), s_\alpha(\beta))} (s_\alpha(\beta), -) = (s_\alpha(\beta))^\vee \in R \quad (2.5)$$

Hence, $R^\vee$ satisfies all requirements for being a root system.

- (2) First, if $t \in E$ is the element used for the polarization of R , then using the element $(t, -) \in E^*$ as the polarization for $R^\vee$, one has the following:

$$\alpha \in R_+ \iff (\alpha, t) > 0 \iff \left(2 \frac{\alpha}{(\alpha, \alpha)}, t\right) > 0 \iff (\alpha^\vee, (t, -)_*) > 0 \iff \alpha^\vee \in (R^\vee)_+ \quad (2.6)$$

Which, the set of positive roots have their coroots being precisely the positive roots of $R^\vee$ under this polarization.

Now, suppose the contrary that $\Pi^\vee$ is not all the simple roots of the coroot system, then there exists $\alpha_i^\vee$ that's not a simple roots.

Let $S = \{r_1^\vee, \dots, r_n^\vee\}$ be the set of simple roots for $R^\vee$ (where each $r_i \in R_+$ since we have the same polarization), this implies that $\alpha_i^\vee = \sum_{j \in J} n_j r_j^\vee$ for some positive integer n_j (Note: Here we eliminate all the zero coefficients of the $r_i^\vee$'s).

Hence, based on the isomorphism between E and E^* , one gets the following:

$$2 \frac{\alpha_i}{(\alpha_i, \alpha_i)} = \sum_{j \in J} n_j \cdot 2 \frac{r_j}{(r_j, r_j)} \implies \alpha_i = \sum_{j \in J} \frac{n_j (\alpha_i, \alpha_i)}{(r_j, r_j)} r_j \quad (2.7)$$

Which, each $\frac{(\alpha_i, \alpha_i)}{(r_j, r_j)} > 0$ because of the positive definiteness of the inner product, together with $0 \notin R$.

Then, since each $r_j \in R_+$, one can write $r_j = \sum_{k=1}^n n_k^j \alpha_k$ where each n_k^j is a nonnegative integer (since $\Pi = \{\alpha_1, \dots, \alpha_n\}$ is the set of simple roots in R_+). Then, we get the following:

$$\alpha_i = \sum_{j \in J} \frac{n_j (\alpha_i, \alpha_i)}{(r_j, r_j)} r_j = \sum_{k=1}^n \sum_{j \in J} \frac{n_j (\alpha_i, \alpha_i)}{(r_j, r_j)} n_k^j \alpha_k \quad (2.8)$$

So, with the linear independence of $\Pi = \{\alpha_1, \dots, \alpha_n\}$, this implies that for $k \in \{1, \dots, n\}$, one has $\sum_{j \in J} \frac{n_j (\alpha_i, \alpha_i)}{(r_j, r_j)} n_k^j = \delta_{ik}$.

And, with each $\frac{n_j (\alpha_i, \alpha_i)}{(r_j, r_j)}$ being positive, and each n_k^j being nonnegative, for $k \neq j$, $\sum_{j \in J} \frac{n_j (\alpha_i, \alpha_i)}{(r_j, r_j)} n_k^j = 0$ implies each $n_k^j = 0$. So, each $r_j = \sum_{k=1}^n n_k^j \alpha_k = n_i^j \alpha_i$, with n_i^j being some nonnegative integer.

Finally, recall that $\alpha \in R$ satisfies $c\alpha \in R$ iff $c = \pm 1$, so the equality $r_j = n_i^j \alpha_i$ enforces $n_i^j = \pm 1$; yet, since $r_j \in R_+$, this further implies $n_i^j = 1$, or each $r_j = \alpha_i$. But, this implies $\alpha_i^\vee = r_j^\vee \in S$ (the set of simple roots of $R^\vee$), which contradicts the assumption that $\alpha_i^\vee$ is not a simple roots.

Hence, our assumption is false, the set $\Pi^\vee = \{\alpha_1^\vee, \dots, \alpha_n^\vee\}$ must be the set of simple roots of $R^\vee$, under this polarization.

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Problem 3

Etingof Problem Set 7.8:

Let $\overline{C_+}$ be the closure of the positive Weyl chamber, and $\lambda \in \overline{C_+}$, $w \in W$ be such that $w(\lambda) \in \overline{C_+}$.

- (1) Show that $\lambda \in \overline{C_+} \cap w^{-1}(\overline{C_+})$.
- (2) Let $L_\alpha \subset E$ be a root hyperplane which separates C_+ and $w^{-1}(C_+)$. Show that then $\lambda \in L_\alpha$.
- (3) Show that $w(\lambda) = \lambda$. Deduce from this that every W -orbit in E contains a unique element from $\overline{C_+}$.

Solution:

- (1) Given that $w(\lambda) \in \overline{C_+}$, it's clear that $\lambda \in w^{-1}(\overline{C_+})$. So by definition $\lambda \in \overline{C_+} \cap w^{-1}(\overline{C_+})$.

- (2) Given L_α being a root hyperplane separating C_+ and $w^{-1}(C_+)$ (and assume that $\alpha \in R_+$), this implies that for any $v \in \overline{C_+}$, one has $(v, \alpha) \geq 0$, and $w^{-1}(v) \in w^{-1}(\overline{C_+}) = \overline{w^{-1}(C_+)}$ satisfies $(w^{-1}(v), \alpha) \leq 0$ (Note: for finite dimensional $\mathbb{R}$ -Euclidean spaces, any linear map is continuous, hence linear isomorphism is a homeomorphism, which preserves closure).

As a result, since $\lambda \in \overline{C_+} \cap w^{-1}(\overline{C_+})$, one has $(\lambda, \alpha) \geq 0$ by the first inequality, while $(\lambda, \alpha) \leq 0$ by the second inequality, so $(\lambda, \alpha) = 0$, showing $\lambda \in L_\alpha$.

- (3) Here we'll pursue by contradiction. Suppose $w(\lambda) \neq \lambda$, let $l = \ell(w)$ (which is the reduced length), here $l > 0$ can be assumed (since for $l = 0$, or $w = \text{id}_E$, the condition $w(\lambda) = \lambda$ is trivial). Then, let $w = s_{\alpha_{i_1}} \circ \dots \circ s_{\alpha_{i_l}}$ be a reduced decomposition into simple roots.

Now, if $l = 1$ (or $w = s_{\alpha_{i_1}}$), then α_{i_1} is a positive root that gets mapped to negative root, showing that $L_{\alpha_{i_1}}$ separates $\overline{C_+}$ and $w^{-1}(\overline{C_+})$, showing $\lambda \in L_{\alpha_{i_1}}$, or $s_{\alpha_{i_1}}(\lambda) = \lambda - \frac{2(\alpha_{i_1}, \lambda)}{(\alpha_{i_1}, \alpha_{i_1})} \alpha_{i_1} = \lambda$ (since $(\alpha_{i_1}, \lambda) = 0$). So, this is another case one needs to eliminate.

Else, if $l > 1$, then there exists positive root β that gets mapped to negative root by w , but not by $s_{\alpha_{i_1}} \circ w$. For instance, choose $\beta = s_{\alpha_{i_l}} \circ \dots \circ s_{\alpha_{i_2}}(\alpha_{i_1})$ does the job, since one has the following:

$$\begin{aligned} w(\beta) &= s_{\alpha_{i_1}} \circ \dots \circ s_{\alpha_{i_l}}(s_{\alpha_{i_l}} \circ \dots \circ s_{\alpha_{i_2}}(\alpha_{i_1})) = s_{\alpha_{i_1}}(\alpha_{i_1}) = -\alpha_{i_1} \\ s_{\alpha_{i_1}} \circ w(\beta) &= s_{\alpha_{i_1}}(-\alpha_{i_1}) = \alpha_{i_1} \end{aligned} \quad (3.1)$$

Then, since $\lambda, w(\lambda) \in \overline{C_+}$, one has $(\lambda, \beta) \geq 0$ (because β is a positive root), while $(w(\lambda), w(\beta)) \leq 0$ (because $w(\beta)$ is a negative root). Yet, since w is an isometry, one has $(\lambda, \beta) = (w(\lambda), w(\beta))$, which the two inequalities enforce $(\lambda, \beta) = 0$. This shows that $s_\beta(\lambda) = \lambda$.

However, it reaches the following equality:

$$\begin{aligned} w(\lambda) &= s_{\alpha_{i_1}} \circ \dots \circ s_{\alpha_{i_l}}(\lambda) = s_{\alpha_{i_1}} \circ \dots \circ s_{\alpha_{i_l}} \circ s_\beta(\lambda) \\ &= (s_{\alpha_{i_1}} \circ \dots \circ s_{\alpha_{i_l}}) \circ ((s_{\alpha_{i_l}} \circ \dots \circ s_{\alpha_{i_2}}) \circ s_{\alpha_{i_1}} \circ (s_{\alpha_{i_2}} \circ \dots \circ s_{\alpha_{i_l}}))(\lambda) \\ &= s_{\alpha_{i_2}} \circ \dots \circ s_{\alpha_{i_l}}(\lambda) \end{aligned} \quad (3.2)$$

Notice that such equality can be used on any possible $\lambda \in \overline{C_+}$ with such property, which in some sense reduces the the expression of w down to a shorter representative in terms of simple reflections, which contradicts the assumption that l is the reduced length of w .

So, one must have $w(\lambda) = \lambda$. This also means that all W -orbit in E has a unique element in $\overline{C_+}$, since if $\lambda \in \overline{C_+}$ and $w \in W$ satisfies $w(\lambda) \in \overline{C_+}$, the above shows that $w(\lambda) = \lambda$. So, it must be unique.

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Problem 4

Etingof Problem Set 7.12:

Let $\varphi : R_1 \xrightarrow{\sim} R_2$ be an isomorphism between irreducible root systems. Show that then φ is a composition of an isometry and a scalar operator: $(\varphi(v), \varphi(w)) = c(v, w)$ for any $v, w \in E_1$.

Solution: It is both sufficient and necessary to prove the case for the simple roots of R_1 (since the simple roots form a basis).

Let $\Pi = \{\alpha_1, \dots, \alpha_n\}$ be the set of simple roots of R_1 under a chosen polarization. Due to the irreducibility of R_1 , there doesn't exist nontrivial proper subset Π_1, Π_2 or Π , such that $\Pi_1 \sqcup \Pi_2 = \Pi$ and $\Pi_1 \perp \Pi_2$.

Now, since φ is an isomorphism of vector spaces, each $\varphi(\alpha_i) \neq 0$, and $(\varphi(\alpha_i), \varphi(\alpha_i)) > 0$. Let us define the value $c_i := \frac{(\varphi(\alpha_i), \varphi(\alpha_i))}{(\alpha_i, \alpha_i)} > 0$ for each index i . Then, since φ preserves each $n_{\beta\alpha}$ for all $\alpha, \beta \in R$, for each indices i, j , one has the following:

$$\frac{2(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)} = n_{\alpha_j \alpha_i} = n_{\varphi(\alpha_j) \varphi(\alpha_i)} = \frac{2(\varphi(\alpha_i), \varphi(\alpha_j))}{(\varphi(\alpha_i), \varphi(\alpha_i))} = \frac{2(\varphi(\alpha_i), \varphi(\alpha_j))}{c_i(\alpha_i, \alpha_i)} \quad (4.1)$$

So, one gets the following equality:

$$c_i(\alpha_i, \alpha_j) = (\varphi(\alpha_i), \varphi(\alpha_j)) \quad (4.2)$$

Which, since i, j are arbitrary, switching the two indices, one gets $c_j(\alpha_i, \alpha_j) = (\varphi(\alpha_i), \varphi(\alpha_j))$ also (since over $\mathbb{R}$ -inner product space, inner product is symmetric). Hence, one gets $c_i(\alpha_i, \alpha_j) = (\varphi(\alpha_i), \varphi(\alpha_j)) = c_j(\alpha_i, \alpha_j)$.

Then, we claim that the collection of c_i 's only has one value (i.e. all c_i 's are the same).

Suppose the contrary that the c_i 's have more than 1 value, then partition Π using the values of c_i : Let Π_1 denotes all simple roots with $c_i = c_1$, Π_2 denotes all simple roots with $c_j \neq c_1$. Then, up to reordering, one can say $\Pi_1 = \{\alpha_1, \dots, \alpha_i\}$ and $\Pi_2 = \{\alpha_{i+1}, \dots, \alpha_n\}$ (which, the assumption provides that both $\Pi_1, \Pi_2 \neq \emptyset$).

For any index $1 \leq k \leq i$ and $i+1 \leq l \leq n$, one has the following:

$$c_1(\alpha_k, \alpha_l) = c_k(\alpha_k, \alpha_l) = c_l(\alpha_k, \alpha_l) \implies (c_1 - c_l)(\alpha_k, \alpha_l) = 0 \quad (4.3)$$

Since $\alpha_l \in \Pi_2$, one has $c_l \neq c_1$, so $(c_1 - c_l) \neq 0$, then the above equality enforces $(\alpha_k, \alpha_l) = 0$, or the two vectors are orthogonal. Yet, with $1 \leq k \leq i$ and $i+1 \leq l \leq n$ being arbitrary, this implies $\Pi_1 \perp \Pi_2$, which contradicts the fact that R_1 is irreducible.

Therefore, all c_i 's must be the same, denote as $c := c_i$ for all i .

Finally, the above equality $c_i(\alpha_i, \alpha_j) = (\varphi(\alpha_i), \varphi(\alpha_j))$ implies $c(\alpha_i, \alpha_j) = (\varphi(\alpha_i), \varphi(\alpha_j))$ for all indices i, j , then φ is a composition of isometry and scalar operator on the simple roots Π , hence on the whole vector space R_1 , and this completes the proof.

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Problem 5

Etingof Problem Set 7.15:

Let $R \subset E$ be an irreducible root system. Show that then E is an irreducible representation of the Weyl group W .

Solution: Suppose the contrary that E is not an irreducible representation of the Weyl group W , which implies some nonzero proper subspace $U \subsetneq E$ is invariant under the Weyl group W 's action. On the other hand, after picking some polarization, let $\Pi = \{\alpha_1, \dots, \alpha_n\}$ be the set of simple roots.

First, pick any nonzero $x \in U$, there must exist $\alpha_i \in \Pi$, such that $(\alpha_i, x) \neq 0$. Since if all $(\alpha_i, x) = 0$, one has $x \in (\text{span } \Pi)^\perp = E^\perp = 0$, which is a contradiction (since Π spans the whole space, while x is assumed to be nonzero). WLOG, say α_i is α_1 up to some reordering.

Then, since x is invariant under the Weyl group action, one has $s_{\alpha_1}(x) = x - \frac{2(\alpha_1, x)}{(\alpha_1, \alpha_1)}\alpha_1 \in U$, showing that $\frac{2(\alpha_1, x)}{(\alpha_1, \alpha_1)}\alpha_1 = x - s_{\alpha_1}(x) \in U$. With $(\alpha_1, x) \neq 0$, one has $\alpha_1 \in U$.

Now, let $\alpha_1, \dots, \alpha_i \in \Pi$ be all the simple roots, such that there exists nonzero $x_l \in U$ satisfying $(\alpha_l, x_l) \neq 0$ (for all index $1 \leq l \leq i$). The above process implies that each $\alpha_l \in U$, hence one has $\text{span}\{\alpha_1, \dots, \alpha_i\} \subseteq U \subsetneq E$, so by linear independence of the α_l 's, $i < n$ (since it must necessarily have a smaller dimension).

However, for all index $i < k \leq n$, since α_k is not in the given list, it implies for all nonzero $x \in U$, it must have $(\alpha_k, x) = 0$. In particular, all index $1 \leq l \leq i$ satisfies $(\alpha_k, \alpha_l) = 0$. So, take the set $\Pi_1 = \{\alpha_1, \dots, \alpha_i\}$ and $\Pi_2 = \{\alpha_{i+1}, \dots, \alpha_n\}$, these two (nonempty) proper subsets of Π are mutually orthogonal by the above claim. Yet, this contradicts the irreducibility of R .

Hence, the assumption must be false, E must be an irreducible representation of the Weyl group W .